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Potential is "intrinsic energy of inertia" label

$$\nabla \cdot \epsilon_0 \mathcal{E} = \varrho \qquad \nabla \cdot \mathcal{B} = 0 \quad \nabla \times \mathcal{E} = -\frac{\partial}{\partial t}\mathcal{B} \qquad \nabla \times \frac{1}{\mu_0}\mathcal{B} = j + \epsilon_0 \frac{\partial}{\partial t}\mathcal{E} \tag{1}$$

$\Rightarrow$

$$\Delta \cdot U - \frac{1}{c^2}\frac{\partial^2}{\partial t^2}U = J \tag{2}$$

$\Rightarrow$ Potential by "ignition J" (cause, inhomgenity) from source at rest, v = 0:

$$U = -\frac{1}{4\pi}\int \frac{J}{|r|}\, dr' \tag{3}$$

elec. Potential structure: $J = -\frac{1}{\epsilon_0}\varrho$

$$U_e := \Phi \overset{\text{elec. charge dens.}=\varrho}{=} \frac{1}{4\pi}\frac{1}{\epsilon_0}\int \frac{\varrho}{|r|}\, dr' \,\hat{=}\, \frac{1}{2}\frac{1}{2\pi}\frac{1}{|r|}\frac{1}{\epsilon_0}Q = \frac{1}{2}\frac{1}{2\pi}\frac{1}{|r|}\frac{1}{\epsilon_0}It \tag{4}$$

magn. Potential structure: $J = -\mu_0 j$

$$U_m := A \overset{\text{elec. current dens.}=j}{=} \frac{\mu_0}{4\pi}\int \frac{j}{|r|}\, dr' \,\hat{=}\, \frac{1}{2}\frac{1}{2\pi}\frac{1}{|r|}\mu_0 I = \frac{1}{2}\frac{1}{2\pi}\frac{1}{|r|}\frac{1}{c^2\epsilon_0}I = \frac{1}{c^2}\frac{1}{t}U_e \overset{\text{source in uniformal movement},v=const.}{=} \tag{5}$$

Electric endogen energy $E_{pot} = mc^2$:

$$e = \text{electron charge} \qquad q = ne = \text{charge} \qquad n \in \mathbb{N} \tag{6}$$

$$\text{Potential} \qquad U = \frac{E_{pot}}{q} = \frac{\left(mc^2\right)^2}{ne} = \frac{m}{q}mc^4 = \kappa mc^4 \tag{7}$$

Electric induced energy $E_{kin} = \frac{1}{2}mv^2 \,\hat{=}\, (pc)^2$:

E_{kin} Energy density electric:

$$w_e = \frac{1}{2}\epsilon_0 \mathcal{E}^2 \tag{8}$$

E_{kin} Energy density magnetic:

$$w_m = \frac{1}{2}\frac{1}{\mu_0}\mathcal{B}^2 \tag{9}$$

E_{kin} Electric Energy:

$$W_e = \frac{1}{2}CU^2 \tag{10}$$

E_{kin} Magnetic Energy:

$$W_m = \frac{1}{2}LI^2 \tag{11}$$

Power:

$$P = VI = (U_1 - U_2)\frac{Q}{t} = \frac{E_1 - E_2}{t} = \frac{W}{t} = \frac{Fr}{t} = Fv \tag{12}$$

Conductor (charge conductive element for energy transfer) condition:

$$\sigma \cdot t \gg \epsilon_0 \tag{13}$$

Slow changing current (quasi-stationary, low frequencies) condition:

$$\frac{v}{c} \ll \frac{1}{c}\frac{\mathcal{E}}{\mathcal{B}} \ll \frac{c}{v} \tag{14}$$

Energy conservation for (large extended) conductive medium:

$$\int \left(\frac{\partial}{\partial t}(we + w_m) + \ \cdot \mathcal{S} + j \cdot \mathcal{E} \right) dr = \frac{d}{dt}W_e + \frac{d}{dt}W_m + D_S + \int (j \cdot \mathcal{E})\, dr \overset{\text{low frequency}}{=} \frac{1}{2}\frac{d}{dt}(LI^2) + \frac{1}{2}\frac{d}{dt}\left(\frac{Q^2}{C}\right) \tag{15}$$

Energy of thin conductor (concentrated conductive medium without electric charge in its surrounding), Voltage (potentialdifference, tension):

$$V = U_1 - U_2 = r\mathcal{E} = \dot{\Psi} + RI + \frac{I}{C}t = L\dot{I} + RI + \frac{Q}{C} \tag{16}$$

$\Delta\cdot :=$ div grad $=$ Laplacian $\quad J =$ "ignition of inhomogenity,cause" $\quad j =$ Current density $\quad \varrho =$ Charge density

$L =$ Inductance $\quad C =$ Capacitance $\quad Q =$ charge (ion) $\quad \Psi = LI =$ mag. Flux

$\mathcal{S} = \mathcal{E} \times \mathcal{H} =$ Poynting-Vector $=$ radiation intensity

$$D_S = \int (\,\cdot\,\mathcal{S})\;dr = \int \mathrm{div}(\mathcal{E}\times\mathcal{H})\,dr = \text{dissipation engergy lost from radiation} \approx 0 \text{ for low frequencies } \omega$$

$$d = \sqrt{\frac{1}{\mu\sigma}\frac{2}{\omega}} = \text{skin depth (surface) of conductor} \qquad r = \frac{d}{\sqrt{2i}}\rho = \frac{d}{1+i}\rho = \text{radius (extension) of conductor}$$

$$\rho = kr = \text{"radius of curvature } k\text{"} \qquad \omega = 2\pi\nu = k\lambda\nu = \frac{1}{\sqrt{LC}} = \text{frequency}$$

$$\mu = \text{permeability of conductor} \qquad \sigma = \text{conductivity of conductor}$$

Low Frequency:

$$\omega \ll \frac{1}{\mu\sigma}\frac{2}{r^2} \iff r \ll d \iff \rho \ll 1 \tag{17}$$
$$L \approx const. \qquad C \approx const. \qquad D_S \approx 0$$

High Frequency:

$$\omega \gg \frac{1}{\mu\sigma}\frac{2}{r^2} \iff r \gg d \iff \rho \gg 1 \tag{18}$$
$$L(t) \neq const. \qquad C(t) \neq const. \qquad \text{for Dipol: } D_S \propto \omega^4 \gg 0$$

Energy:

Action Quantum, Planck Constant (Energy-Frequency-Slope)[1]:

$$\begin{gathered} h = \tan(\alpha) = \frac{E_{kin}}{\nu_i - \nu_0} = \frac{eV_{0i}}{\nu_i - \nu_0} \equiv \\ \frac{E_1 - E_2}{\nu_0} = \frac{W}{\nu_0} = \frac{F_{EM}\cdot x}{\nu_0} = \hbar k\lambda \end{gathered} \tag{19}$$

[1] Wirkungsquantum

$$\hbar = \frac{E_1 - E_2}{\omega} = \tag{20}$$

$$E_{kin}^{max} = eV_0 = e\frac{Q}{C} = h(\nu - \nu_0) \tag{21}$$

$$V_0 = \text{Opposing Potential } (I = 0)$$

$$\alpha = \angle(E_{kin}, \nu) \qquad e = \text{electron charge} \qquad Q = \text{ion charge}$$

$$C = \text{Capacitance} \qquad \nu = \frac{1}{2\pi}\frac{1}{\sqrt{LC}}$$

$$\Rightarrow \text{Inductance } L = \frac{4\pi^2}{C}\left(\frac{e}{h}Q + \nu_0\right)^2$$

$\Rightarrow$ Ionisation Energy of chemical element $E_i = h\nu = h\frac{\lambda}{c} = \hbar\omega = \frac{h}{k\lambda}\omega$

$\Rightarrow$ Engergy released (gained) $E_{gain} = E_{pot} - E_i = mc^2 - h\nu$